Does a Pending-Sector Warning Predict Hard-Drive Failure? A Survival Analysis of Backblaze Drive Stats, Q1 2026

Advanced Statistical Methods II — Group Project (Type I)

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Abstract

We studied whether a SMART “pending sector” warning (SMART 197, Current Pending Sector Count) on a fixed day is associated with a later recorded hard-drive failure. We used a survival framework on Backblaze’s public Drive Stats data.

- **Cohort:** 18,533 Seagate ST12000NM0008 drives in service on 1 January 2026, followed by calendar day to 31 March 2026.
- **Methods:** Kaplan–Meier curves by baseline SMART 197 status and a Cox proportional-hazards model. Exits and observation gaps get pre-specified sensitivity analyses, and a secondary model adjusts for powered-on age.
- **Results:**
  - Among the 162 drives with a nonzero baseline count, 28 had a recorded failure within 89 days. The estimated proportion without a recorded failure was 0.826 (95% CI 0.758–0.877).
  - Among the 18,371 drives with a zero count, it was 0.9955 (0.9944–0.9964), with 83 failures.
  - The Cox hazard ratio was **43.2 (95% CI 28.1–66.3)**, so we reject the null hypothesis of no association (Wald $p \approx 10^{-66}$).
- **Robustness:** the result holds when exits are reclassified (HR 43.2), when gap days are removed (44.1), and when adjusting for powered-on years (43.6). Hazards were not proportional: most of the excess risk came in the first month.
- **Caveat:** Backblaze counts proactive replacements triggered by SMART indicators as failures. The association is therefore partly built into the outcome. It should not be read as a causal effect or as a validated prediction rule.

1. Introduction and hypotheses

Hard drives report self-monitoring (SMART) attributes, and data centres use them to decide when to replace drives. SMART 197 counts sectors the drive could not read and has marked as “pending” for reallocation. Our question, as approved in our project proposal (6 October 2026), was:

> *Is a pre-specified SMART health indicator associated with subsequent hard-drive failure, after accounting for drive age?*

- **H_0:** the baseline SMART 197 status has no association with the subsequent failure hazard.
- **H_a:** the baseline SMART 197 status is associated with the subsequent failure hazard.

Survival analysis fits this question because drives are observed for different lengths of time. Many drives never fail in the window (right censoring), and some leave observation for reasons other than failure.

2. Background

- **SMART signals and failure.** Pinheiro, Weber and Barroso [6] studied more than 100,000 drives at Google.
  - They found that a few SMART parameters "have a large impact on failure probability", including probational (pending) counts.
  - After the first probational count, drives were "16 times more likely to fail within 60 days" than drives with none.
  - They also found that over 56% of failed drives had no count in the four strongest SMART signals, so SMART-only models are "unlikely to be useful for predicting individual drive failures".
- **How Backblaze defines failure.** In Drive Stats a drive is recorded as failed when it is removed and replaced. That covers *reactive* failures, where the drive dies or stops responding. It also covers *proactive* failures, where "indicators such as SMART stats, FSCK checks, etc., signal that the drive is having difficulty and drive failure is highly probable" [7].
  - The public data records only a 0/1 failure flag, not its type [7].
  - In Q3 2024, 721 of Backblaze's 1,361 failures (53%) were proactive [7].
  - The Q3 2025 report describes failure monitoring through SMART attributes, including `Current_Pending_Sector` [8].

  This matters for interpretation (§7).

3. Data

Source. Backblaze Drive Stats, Q1 2026: 90 daily CSV files, 1 January to 31 March 2026, 30.6 million drive-days in total [9]. The archive's SHA1 and every file's CRC32 match the published values. No raw data is stored in the project repository.

Choice of drive model. We restricted the study to one model, Seagate `ST12000NM0008` (12 TB), so that manufacturer and firmware differences don't confound the comparison. Q1 has 18,650 such drives and 129 failure flags.

Cohort construction. Baseline is 1 January 2026 (day 0).

- **Included:** drives present on that day.
- **Excluded:** drives first seen later (116 drives, 17 of them with failures) and one drive whose failure flag fell on the baseline day.

That leaves **18,533 drives and 111 recorded failures**. Data checks:

- There are no duplicate drive-day records, and no rows after a failure.
- Baseline SMART 197, SMART 5 and SMART 9 were present for every included drive.

Outcome and censoring.

- **Event:** the first `failure = 1` flag after baseline, dated on the flag day.
- **Censoring:**
  - Drives still present on 31 March are censored there.
  - 96 drives disappeared earlier without a flag. They are censored at their last observed day and never counted as failures.
- **Gaps:** 10.2% of drives have at least one gap in observation. The share is 22% among SMART 197 > 0 drives and 10% among SMART 197 = 0 drives.

Exposure. SMART 197 raw > 0 versus = 0, taken from the 1 January record only: 162 drives versus 18,371. Later values are never used, so the exposure cannot "see the future".

Table 1. Cohort accounting (included drives).

Baseline SMART 197	Drives	Recorded failures	Censored 31 Mar	Unexplained exits
= 0	18,371	83	18,195	93
> 0	162	28	131	3
Total	18,533	111	18,326	96

4. Methods

Time scale. Calendar days since 1 January 2026, from 0 to 89. This is a prevalent cohort. Drives entered service before 2026, and their installation dates are unknown.

Kaplan–Meier estimator [1], with 95% pointwise confidence intervals from Greenwood's variance [3] on the log(−log) scale. We report survival at days 30, 60 and 89.

Cox proportional-hazards model [2]: hazard $h(t \mid x) = h_0(t) \cdot \exp(\beta x)$, where x indicates baseline SMART 197 > 0.

- Tied failure days are handled with Efron's method [4].
- We test H_0: $\beta = 0$ with the Wald test.
- Proportional hazards are checked with the scaled-Schoenfeld-residual test of Grambsch and Therneau [5]. The test is reported, but it was not used to change the model.

Pre-specified sensitivity analyses.

- **S1 (exits).** A bounded check of 1–14 April 2026 found that 93 of the 96 exits reappeared. In S1 those drives are censored at day 89 instead of their last Q1 day.
- **S2 (gaps).** Days a drive was missing are removed from the risk set, using start–stop (counting-process) intervals.

Secondary analysis: adjustment for age. Drive installation dates are not published. SMART 9 counts powered-on hours, so we used baseline SMART 9 ÷ 8,766 as "powered years", a proxy for age. We fitted one extra Cox model with SMART 197 status plus a

linear powered-years term. It was specified and committed before fitting, and added after the primary plan to honour the “after accounting for drive age” in our proposal.

Safeguards against data-driven choices. Before anyone looked at failure counts by group, we committed to the repository:

- the decision rule for using SMART 197 rather than SMART 5 (keep 197 if its nonzero group had at least 10 failures);
- the censoring and gap conventions;
- the full analysis plan.

Software. Python 3.13, lifelines 0.30.0 [10] and statsmodels 0.15.0 [11].

- Every Cox coefficient was cross-checked between the two packages and agreed to within 10^{-6}.
- Damped Newton steps were needed, because the rare exposure and strong effect made full steps from $\beta = 0$ overflow.

5. Results

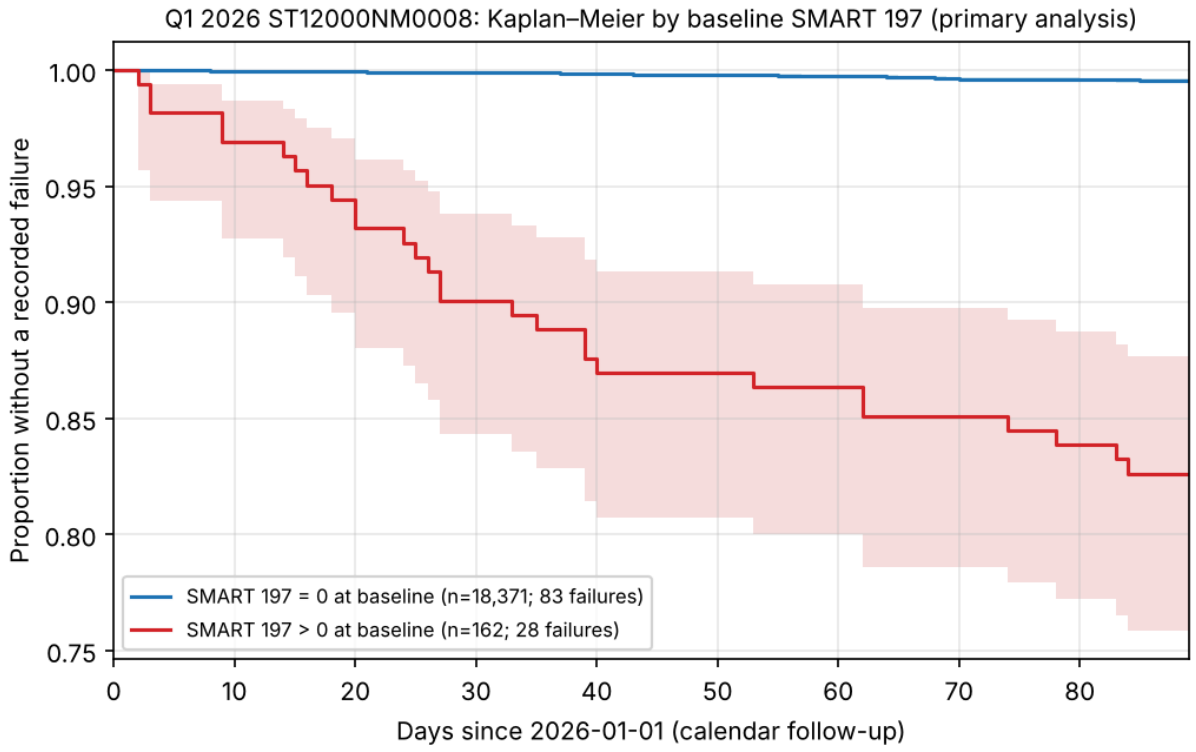


Figure 1. Kaplan–Meier estimates of the proportion of drives without a recorded failure, by baseline SMART 197 status (shaded: 95% pointwise CI).

Table 2. Kaplan–Meier estimates (primary analysis), with 95% CI.

Group	Day 30	Day 60	Day 89
SMART 197 = 0	0.9989 (0.9982–0.9993)	0.9976 (0.9968–0.9982)	0.9955 (0.9944–0.9964)
SMART 197 > 0	0.9007 (0.8431–0.9380)	0.8635 (0.8001–0.9079)	0.8262 (0.7583–0.8766)

Drives at risk on days 0, 30, 60 and 89:

- SMART 197 = 0: 18,371; 18,349; 18,324; 18,195
- SMART 197 > 0: 162; 145; 139; 131

Table 3. Cox models: hazard ratio for baseline SMART 197 > 0 versus = 0.

Analysis	HR	95% CI	Wald p	PH test p
Primary	**43.2**	28.1–66.3	1.7×10^{-66}	0.0069
S1: exits reclassified	43.2	28.1–66.3	1.7×10^{-66}	0.0069
S2: gap days removed	44.1	28.7–67.7	3.5×10^{-67}	—
Adjusted for powered years	43.6	28.4–66.9	9.5×10^{-67}	0.0070

S2 has no PH test, because lifelines provides none for start–stop fits.

Age-adjusted model.

- Powered years have a hazard ratio of 1.11 per year (95% CI 0.85–1.43, p = 0.44), with no PH violation (p = 0.92).
- Almost all drives are 5.5–6.2 powered years old. The median is 5.73 in the zero group and 6.00 in the nonzero group.

Decision on the hypotheses. At the 5% level we reject H_0: baseline SMART 197 status is associated with the subsequent hazard of a recorded failure. This holds both without and with adjustment for powered-on age.

Shape of the effect. About 10% of drives with SMART 197 > 0 had a recorded failure within 30 days, and about 4% in each later 30-day span. The PH test rejects a constant hazard ratio (p = 0.007). The HR of 43 is therefore an average over the quarter, larger early and smaller later, and the Kaplan–Meier curves are the more faithful summary.

In absolute terms. About 17 in 100 drives with a pending-sector warning had a recorded failure within three months, against fewer than 1 in 200 without one. The warning group is under 1% of the fleet, but it accounts for 25% of the failures (28 of 111).

6. Discussion

The direction and size of the association are in line with Pinheiro et al. [6]. They found that drives with probational counts were 16 times more likely to fail within 60 days. Our larger hazard ratio may reflect several differences:

- the different outcome definition;
- the strong early hazard;
- one drive model;
- a fixed-baseline comparison rather than “after the first event”.

The result is robust to every pre-specified sensitivity analysis and to adjustment for powered-on age. Our exit check showed that 93 of 96 apparent exits were temporary gaps in the data, not departures from the fleet.

7. Limitations

1. **The outcome is partly defined by SMART signals.** Backblaze records proactive, SMART-informed replacements as failures [7], and the public flag doesn't say which kind. Part of the association therefore reflects the operator's replacement policy acting on warnings such as SMART 197, not spontaneous mechanical failure. The two can't be separated with these data.
2. **Prevalent cohort, left truncation, no true age.** Installation dates are unknown. Powered-on hours are a proxy, and they vary over a narrow range here, so the age adjustment rules out confounding by age only within that range.
3. **Recording of failure dates.** Dates are daily snapshot dates. 21 of the 44 failures that followed a gap were flagged on the day the drive reappeared, so recorded dates may lag true failure.
4. **Informative exits.** 4 drives that left Q1 carried a failure flag in early April.
5. **Overlap with other warnings.** 110 of the 162 SMART 197 > 0 drives also had reallocated sectors (SMART 5 > 0). The HR is not specific to pending sectors.
6. **Scope.** One drive model, one quarter and 28 exposed failures, hence wide confidence intervals. No other covariates were available or planned.

8. Conclusion

For Seagate ST12000NM0008 drives in service on 1 January 2026, a nonzero pending-sector count marked a small group, under 1% of drives, with a far higher three-month rate of recorded failure: HR 43.2, or 43.6 after adjustment for powered-on age. The excess was concentrated in the first weeks and was robust to how exits, gaps and age were handled. Because Backblaze counts SMART-informed replacements as failures, this is best stated as a strong association between SMART 197 and recorded removal-or-failure. It is not evidence that pending sectors cause failure, nor a validated prediction rule.

References

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3. Greenwood, M. (1926). The natural duration of cancer. *Reports on Public Health and Medical Subjects*, 33, HMSO.
4. Efron, B. (1977). The efficiency of Cox's likelihood function for censored data. *JASA*, 72, 557–565.
5. Grambsch, P. M. & Therneau, T. M. (1994). Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*, 81, 515–526.
6. Pinheiro, E., Weber, W.-D. & Barroso, L. A. (2007). Failure trends in a large disk drive population. *Proc. 5th USENIX FAST*, 17–28.
7. Backblaze (2024). Backblaze Drive Stats for Q3 2024. https://www.backblaze.com/blog/backblaze-drive-stats-for-q3-2024/
8. Backblaze (2025). Backblaze Drive Stats for Q3 2025. https://www.backblaze.com/blog/backblaze-drive-stats-for-q3-2025/

9. Backblaze. Hard Drive Data and Stats (Q1 2026 archive). https://www.backblaze.com/cloud-storage/resources/hard-drive-test-data
10. Davidson-Pilon, C. (2019). lifelines: survival analysis in Python. *Journal of Open Source Software*, 4(40), 1317.
11. Seabold, S. & Perktold, J. (2010). statsmodels: econometric and statistical modeling with Python. *Proc. 9th Python in Science Conference*.

Appendix: reproducibility

All code, tests, decisions and outputs are in the project repository `aayush-0131/backblaze-survival-analysis`. Raw and drive-level data are never committed.

Order of the safeguards (commit history):

- `e3cf5dd`: decision rules committed before the per-group counts were read.
- `4259dc3`: per-group counts opened.
- `2b91f38`: primary analysis plan committed before fitting.
- `28cb715`: primary results.
- `e117f03`: age-adjusted plan committed before fitting.
- `714fb97`: age-adjusted results.

The cohort was rebuilt independently from a freshly downloaded, hash-verified archive and reproduced byte for byte. The test suite (49 tests) passes.

To reproduce, download the Q1 ZIP to `data/raw/`, then run:

```
PYTHONPATH=src python scripts/build_cohort_skeleton.py --zip-path data/raw/data_Q1_2026.zip --
expect-manifest results/audit/data_manifest.json --output-dir data/processed/q1_cohort
PYTHONPATH=src python scripts/fit_q1_age_adjusted.py --check
PYTHONPATH=src python scripts/fit_q1_survival.py --zip-path data/raw/data_Q1_2026.zip --check
```

The last command also needs the April exit-linkage output (`scripts/link_q1_exits.py`).